

Answer Key

Lesson: 1

- 1 scene 2 geometry, material 3 material
- 4 renderer 5 NormalMap
- 6 The 'package.json' file contains information about the dependencies and packages required for the project.
- 7 The command `npm init -y` creates the package.json file.
- 8 The `npm install` command creates the nodes modules folder.
- 9 The `npm dev run` command is used to run and test the code.
- 10 A normal map/texture is an image that creates the illusion of surface details like bumps, dents, and grooves without modifying the actual 3D model's geometry.
- 11 False 12 True 13 False
- 14 True 15 False

Lesson: 3

- 1 c. React 2 a. It is a JavaScript library
- 3 b. App.js 4 c. <App> 5 `npm start`
- 6 grid 7 Route

Lesson: 2

- 1 Environment maps 2 GLTF/GLB
- 3 Graphics Library Transmission Format
- 4 compact 5 load() 6 texture
- 7 environment 8 False 9 False 10 True
- 11 False 12 True
- 13 Before using the GLTF/GLB format for importing models, the GLTFLoader package from Three.js must be imported.
- 14 'gltf.scene' holds the complete hierarchy of objects, meshes, and materials that make up the loaded model.
- 15 The two classes that can be used to apply an equirectangular map to the background are TextureLoader class and RGBELoader class.
- 16 TextureLoader is used for standard images, such as JPEG or PNG files while RGBELoader is used for high dynamic range (HDR) images, typically in .hdr or .exr formats, to capture a wider range of lighting information for realistic reflections and lighting effects.

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HDR (High Dynamic Range)		JPG (Low Dynamic Range)
High (Stores bright and dark details)	 Dynamic Range	Low (Limited brightness range)
.hdr, .exr	 File Format	.jpg, .png
Realistic reflections and illumination	 Lighting Quality	Flat reflections, less realism
Higher file size, requires more GPU Power	 Performance	Lightweight, loads faster
RGBE Loader	 Loader in Three.js	Texture Loader

Lesson: 4

- 1 b. *CREATE TABLE*
- 2 b. *AUTO_INCREMENT*
- 3 a. *npm install*
- 4 c. ?
- 5 *app.listen()*
- 6 mysql2
- 7 Cross Origin Resource Sharing
- 8 *useState*
- 9 False
- 10 True

Lesson: 6

- 1 b. The force needed to adjust a vehicle's movement
- 2 c. *createVector()*
- 3 b. To allow agents to move intelligently and respond dynamically
- 4 d. Teleport
- 5 b. Moves away from a target
- 6 c. Target position – Current position
- 7 b. The vehicle slows down as it reaches a target
- 8 True
- 9 True

Lesson: 8

- 1 Computer Vision
- 2 Pre-trained
- 3 Grayscale
- 4 True
- 5 True
- 6 False

Lesson: 5

- 1 Joins
- 2 *CREATE USER*
- 3 *IDENTIFIED BY*
- 4 *mysql.user*
- 5 *FLUSH PRIVILEGES*
- 6 a. References
- 7 False
- 8 False
- 9 True
- 10 False
- 11 *REVOKE privilege ON database_name.table_name FROM 'username'@'hostname';*

Lesson: 7

- 1 Statistics
- 2 R^2 Score
- 3 Machine learning
- 4 False
- 5 False
- 6 True
- 7 True